

CSA 0378-DATA STRUCTURES

21. QUICK SORT

```
main.c
1 #include <stdio.h>
2
3 int partition(int a[], int low, int high) {
4     int pivot = a[high];
5     int i = low - 1, j, temp;
6
7     for(j = low; j < high; j++) {
8         if(a[j] < pivot) {
9             i++;
10            temp = a[i];
11            a[i] = a[j];
12            a[j] = temp;
13        }
14    }
15
16    temp = a[i+1];
17    a[i+1] = a[high];
18    a[high] = temp;
19
20    return i+1;
21 }
22
23 void quicksort(int a[], int low, int high) {
24     int pi;
25
26     if(low < high) {
27         pi = partition(a, low, high);
28         quicksort(a, low, pi-1);
29         quicksort(a, pi+1, high);
30     }
31 }
32
33 int main() {
34     int a[] = {50,20,80,10,40};
35     int n = 5, i;
36
37     quicksort(a,0,n-1);
38
39     printf("Sorted Array: ");
40     for(i=0;i<n;i++)
41         printf("%d ",a[i]);
42
43     return 0;
44 }
```

Output

Sorted Array: 10 20 40 50 80

*** Code Execution Successful ***